Energy Management for ZED Certification

Practical Workshop — Unnathi CNC Technologies Pvt Ltd

By Akshaya Createch

SECTION 1: ZED & Energy Management — The Big Picture

What is ZED Certification?

Zero Defect Zero Effect (ZED) — A Government of India initiative for manufacturing MSMEs

- **Zero Defect** — Products meet quality standards consistently
- **Zero Effect** — Manufacturing has minimal environmental impact

Certification Levels:

Level	Parameters	Assessment
Bronze	Basic compliance across 20 parameters	Self-assessment + desk review
Silver	Intermediate compliance with evidence	On-site assessment
Gold	Advanced compliance with demonstrated results	Rigorous on-site assessment

> ***Key Point:*** *Energy Management is* ***Parameter 12*** *— one of 20 ZED parameters, and it directly impacts your certification level.*

Parameter 12: Energy Management — The 5 Requirements

#	Requirement	What the Assessor Checks
12.1	All energy sources identified & targets set	Energy source register, target documents, policy
12.2	Consumption measured & recorded; corrective actions on deviations	Consumption records, deviation reports, corrective actions taken
12.3	Employees trained in energy management	Training records, attendance sheets, effectiveness assessment
12.4	Periodic energy audits in place	Internal audit reports, findings, follow-up actions
12.5	RCA & CAPA processes in place	Root cause analysis records, CAPA closure evidence

> ***The assessor doesn't just check if documents exist — they check if the system is LIVING and ACTIVE.***

How Energy Management Connects to Other ZED Parameters

Energy management doesn't exist in isolation. The skills you learn today apply across ZED:

Skill	Energy (12)	Also Used In
Target setting & tracking	Energy targets	Quality (5), Measurement (14)
Internal audits	Energy audits	Environment (13), Safety (3), Quality (5)
RCA / CAPA	Energy deviations	ALL parameters (5, 8, 9, 10, 12, 13, 14)
Training & awareness	Energy training	Safety (3), Quality (5), Environment (13)
Management review	Energy review	All — through MRM

> ***Learn it once for energy, apply it everywhere for ZED.***

SECTION 2: Your Energy Profile — Understanding the Numbers

Unnathi CNC — Energy Sources

Energy Source	Type	Used For
Grid Electricity (EB)	Primary	CNC machines, lighting, HVAC, pumps, ancillary equipment
Solar PV	Renewable	Supplements grid electricity
Compressed Air	Secondary	Tool changes, work holding, cleaning
Hydraulics	Secondary	Powering clamps and equipment

Your Equipment (Major Energy Consumers):

- 9 CNC Machining Centers
- 19 CNC Turning Centers
- 3 Grinding Machines
- Air Compressors, HVAC, Pumps, Coolant systems

Your Solar Power Production (Jul-24 to Jun-25)

Monthly Target: 8,000 kWh | Annual Target: 5% increase yearly

Month	Production (kWh)	vs Target	Status
Jul-24	9,136	+1,136	Above
Aug-24	7,498	-502	**Below**

Sep-24	8,572	+572	Above
Oct-24	9,110	+1,110	Above
Nov-24	8,060	+60	Above
Dec-24	6,967	-1,033	**Below**
Jan-25	7,456	-544	**Below**
Feb-25	8,815	+815	Above
Mar-25	8,934	+934	Above
Apr-25	10,866	+2,866	Above
May-25	10,301	+2,301	Above
Jun-25	9,551	+1,551	Above

Annual Total: ~1,05,266 kWh | Monthly Average: ~8,772 kWh

> *9 out of 12 months exceeded target. 3 months below due to weather (rain/cyclone).*

Your Grid Power Consumption (Jul-24 to Jun-25)

Target: 10% reduction yearly

Month	Consumption (kWh)	Trend
Jul-24	13,683	Lowest
Aug-24	16,360	
Sep-24	16,169	
Oct-24	14,929	
Nov-24	15,999	
Dec-24	17,414	
Jan-25	20,645	**Highest**
Feb-25	17,100	
Mar-25	17,683	
Apr-25	19,061	
May-25	19,783	
Jun-25	16,186	

Annual Total: ~2,05,012 kWh | Monthly Average: ~17,084 kWh

The Complete Energy Picture

Metric	Value
Total annual energy	~3,10,278 kWh
Solar contribution	~1,05,266 kWh (**34%**)
Grid contribution	~2,05,012 kWh (66%)
Average monthly total	~25,856 kWh
Peak consumption month	Jan-25 (Grid: 20,645 + Solar: 7,456 = 28,101 kWh)
Lowest consumption month	Jul-24 (Grid: 13,683 + Solar: 9,136 = 22,819 kWh)

Key Observations:

1. **34% renewable energy** — excellent for a manufacturing SME
2. **Inverse relationship** — when solar dips (monsoon), grid consumption rises
3. **Jan-25 spike** — needs investigation (lowest solar + highest grid = most expensive month)
4. **Summer months (Apr-May)** — highest solar but also high grid = highest production period?

> ***Question for discussion:*** *Do your production volumes correlate with these patterns?*

SECTION 3: Systematic Energy Recording & Target Setting

Why Systematic Recording Matters

Manual Register (Current)	Systematic Approach (Needed)
Monthly totals from EB bills	Daily meter readings
No breakdown by section/machine	Section-wise consumption
Difficult to spot daily deviations	Immediate deviation detection
Hard to correlate with production	Production-linked energy data
Assessor sees basic compliance	Assessor sees active management

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Setting Up an Energy Register

What to Record Daily:

Time	Reading Source	Data Point	Who Records
Shift start	Main meter	Opening reading (kWh)	Designated operator

Shift end	Main meter	Closing reading (kWh)	Designated operator
Shift start	Solar inverter	Daily generation (kWh)	Designated operator
Monthly	EB bill	Total units, max demand, PF, bill amount	Accounts/Admin

Frequency of Recording:

Data	Frequency	Purpose
Meter readings	Daily	Trend monitoring, deviation detection
Solar generation	Daily	Renewable performance tracking
EB bill data	Monthly	Cost analysis, target tracking
Section-wise consumption	Weekly/Monthly	Identifying high-consumption areas
Specific Energy Consumption	Monthly	Normalized performance metric

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Setting SMART Energy Targets

Step 1: Calculate Your Baseline

- Use last 12 months average as baseline
- Your grid baseline: **17,084 kWh/month**
- Your solar baseline: **8,772 kWh/month**

Step 2: Set Quarterly Reduction Targets

Quarter	Grid Target (kWh)	Reduction	Solar Target (kWh)	Increase
Q1 (Baseline)	17,084	—	8,772	—
Q2	16,570	-3%	9,210	+5%
Q3	16,073	-3%	9,671	+5%
Q4	15,591	-3%	10,155	+5%

Step 3: Normalize Targets (Advanced)

- **Specific Energy Consumption (SEC)** = Total kWh / Units Produced
- Example: If you produce 5,000 parts/month using 17,084 kWh → SEC = 3.42 kWh/part
- Target: Reduce SEC by 5% → 3.25 kWh/part
- This is a fairer measure because it accounts for production volume changes

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Tracking Deviations

When is it a deviation?

- Consumption exceeds target by **more than 10%** in any month
- Consumption exceeds target for **2 consecutive months**
- Any sudden spike (>20% from previous month)

What to do when deviation is found:

1. **Record** the deviation (what, when, how much)
2. **Analyze** the root cause (see Section 5: RCA/CAPA)
3. **Act** — implement corrective action if needed
4. **Document** everything — this is your ZED evidence

Hands-On Exercise 1

Using your actual data, calculate:

1. Your baseline SEC (you'll need production volume data)
2. A quarterly target for the next 4 quarters
3. Identify which months from Jul-24 to Jun-25 would count as "deviations"

Work in groups of 3-4. Time: 15 minutes.

SECTION 4: Internal Energy Audit — Practical Guide

What is an Internal Energy Audit?

A systematic check of:

- Are we consuming energy efficiently?
- Are our records accurate and up to date?
- Are there energy wastage points?
- Are we meeting our targets?
- What can be improved?

For ZED: You must conduct energy audits periodically (at least quarterly recommended)

Energy Audit Checklist — Section by Section

A. Documentation Check

- [ ] Energy policy displayed and current?
- [ ] Energy source register complete and updated?
- [ ] Consumption records maintained (daily/monthly)?
- [ ] Targets set and tracking in place?
- [ ] Previous audit findings closed?

B. Metering & Monitoring

- [ ] All meters functional and calibrated?
- [ ] Readings being taken as per schedule?
- [ ] Solar inverter readings recorded?
- [ ] EB bill data reconciled with meter readings?

C. CNC Machines (Major Consumers)

- ☐ Machines switched off when idle for >15 minutes?
- ☐ Standby power consumption minimized?
- ☐ Machining parameters (speed, feed) optimized for energy?
- ☐ Coolant pumps running only when needed?
- ☐ No unnecessary machines running during low-production periods?

D. Compressed Air System

- ☐ Air leaks checked and fixed? (Compressed air leaks can waste 20-30% of energy)
- ☐ Compressor running at optimal pressure?
- ☐ Air receiver tank pressure adequate?
- ☐ Blow guns used efficiently (not for cleaning where not needed)?

E. Lighting

- ☐ All LED? (You have this — Yes)
- ☐ Lights off in unoccupied areas?
- ☐ Motion sensors installed? (You don't have this yet — OI)
- ☐ Natural daylight utilized where possible? (You have this — Yes)

F. HVAC

- ☐ Temperature set at energy-efficient levels? (Your OI area)
- ☐ Filters cleaned regularly?
- ☐ Doors/windows closed when AC is running?
- ☐ VFDs functioning properly? (You have this — Yes)

G. General

- ☐ Computers/monitors turned off after work hours?
- ☐ Energy awareness signage displayed? (You have this — Yes)
- ☐ Water pumps running only when needed?

Conducting the Audit — Step by Step

1. **Plan** — Schedule date, assign auditor (should be different from the area being audited)
2. **Prepare** — Print checklist, review previous findings, carry measuring tools if needed
3. **Walk** — Go section by section through the factory
4. **Record** — Note findings with specific details (where, what, how much waste estimated)
5. **Report** — Write audit report with findings, categorized as:
    - **Conformity** — Meeting requirements
    - **Observation** — Minor issue, no immediate risk
    - **Non-Conformity (NC)** — Significant gap requiring CAPA
    - **Opportunity for Improvement (OI)** — Good practice not yet implemented
6. **Follow up** — Track closure of NCs and OIs

Hands-On Exercise 2: Mini Factory Walkthrough

We will walk through one section of the factory and conduct a real mini-audit.

- Groups of 3-4
- Use the checklist provided
- 20 minutes for the walkthrough
- 10 minutes to write up findings
- Each group presents their top 3 findings

SECTION 5: Root Cause Analysis & CAPA for Energy Deviations

Why RCA/CAPA for Energy? (ZED 12.5)

The ZED assessor wants to see that when energy consumption deviates from targets, you:

1. **Investigate** why it happened (Root Cause Analysis)
2. **Fix** it (Corrective Action)
3. **Prevent** it from happening again (Preventive Action)
4. **Document** the entire process

> *This is the **same process** you use for quality, maintenance, safety — "Same RCA procedure" as noted in your ZED documents.*

The 5-Why Method

Simple, effective, and assessor-approved.

Example from Your Data:

Problem: Grid consumption in Jan-25 was 20,645 kWh — highest in 12 months (21% above average)

#	Why?	Answer
1	Why was grid consumption high in Jan-25?	Because solar generation was low (7,456 kWh — below 8,000 target)
2	Why was solar generation low?	Weather conditions — overcast skies, shorter days in winter
3	Why did low solar cause such a big grid spike?	Because production demand remained the same, so grid had to compensate
4	Why wasn't the production schedule adjusted?	Because there was no mechanism to plan production load based on solar availability
5	Why is there no such mechanism?	Energy consumption and production planning are not linked currently

Root Cause: No integration between energy availability (solar forecast) and production scheduling

Corrective Action: For immediate relief — none needed (weather is external factor)

Preventive Action: Consider scheduling energy-intensive operations during high-solar hours (10 AM to 3 PM); review production planning to account for seasonal solar variation

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Example 2: When Corrective Action is NOT Required

Problem: Solar production below target in Aug-24 (7,498 kWh), Dec-24 (6,967 kWh), Jan-25 (7,456 kWh)

Analysis:

- All three months had weather events (monsoon rain, cyclone effect, winter overcast)
- These are **external, uncontrollable** factors
- Production from solar panels is inherently weather-dependent
- 9 out of 12 months exceeded the target

Conclusion: Deviation noted and documented. Root cause is external weather conditions. Corrective action is **not required**. Will continue monitoring.

> ***This is exactly what your records already state*** *— but having it in a formal RCA/CAPA format makes it assessor-ready.*

CAPA Form Template

Field	Details
CAPA No.	CAPA-EN-001
Date Raised	DD/MM/YYYY
Raised By	Name, Department
Category	Energy Deviation
Description	Grid consumption exceeded target by X% in Month/Year
Immediate Action	What was done right away (if anything)
Root Cause (5-Why)	Document the 5-why chain
Corrective Action	What was done to fix this specific occurrence
Preventive Action	What will prevent recurrence
Target Closure Date	DD/MM/YYYY
Responsible Person	Name
Verification	How will we confirm the action worked
Status	Open / In Progress / Closed
Closed By	Name, Date
Reviewed By	Management review (MRM reference)

Hands-On Exercise 3: RCA/CAPA Practice

Scenario: In a hypothetical month, your compressed air system consumed 30% more energy than usual.

- Work in groups

- Use the 5-Why method to find the root cause
- Fill in the CAPA form template
- Present your analysis

Time: 15 minutes

SECTION 6: Digital Energy Management — The Path Forward

Current State vs Desired State

Aspect	Current (Manual)	Desired (Digital)
Data recording	Paper registers, EB bills	Daily digital entry → automated
Target tracking	Annual review from paper records	Real-time dashboard with alerts
Deviation detection	End-of-month when bill arrives	Daily/weekly trend monitoring
Audit trails	Physical files, hard to search	Searchable, timestamped records
Reports for assessor	Manual compilation	One-click generated reports
RCA/CAPA	Paper forms	Tracked workflow with deadlines
Training records	Signed paper sheets	Digital records with certificates

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The Journey: Manual → Digital → IoT

Phase 1: Manual Entry Digital System (Immediate)

- Operators enter daily meter readings into a web-based system
- System calculates targets, flags deviations, generates reports
- All ZED evidence generated automatically
- Works on phone, tablet, or computer — no app installation

Phase 2: Smart Metering (3-6 months)

- Install IoT energy meters at main load points
- Automatic data capture — no manual entry needed
- Real-time monitoring and alerts
- Machine-level energy tracking

Phase 3: Intelligent Energy Management (6-12 months)

- AI-driven optimization suggestions
- Production scheduling based on energy availability
- Predictive maintenance based on energy consumption patterns
- Automated reporting to management

> ***Today's training prepares your team for Phase 1. The system is ready — we'll demonstrate briefly.***

SECTION 7: Assessment Readiness

What the ZED Assessor Will Ask About Energy

Assessor Question	Expected Evidence
"Show me your energy policy"	Printed, signed, displayed policy document
"What are your energy sources?"	Energy source register with details
"How do you track consumption?"	Daily/monthly records (register or system)
"What are your energy targets?"	Documented targets with baseline
"Show me last 6 months consumption vs target"	Tracking sheet or dashboard showing trends
"What do you do when you exceed the target?"	RCA/CAPA records showing analysis and actions
"How do you train employees on energy?"	Training records, attendance, effectiveness
"Show me your last energy audit report"	Internal audit report with findings and follow-up
"What improvements have you made?"	List of energy-saving initiatives with results
"How does management review energy performance?"	MRM minutes showing energy discussion

Your Readiness Scorecard

Requirement	You Have	Gap
Energy Policy (12.1)	Yes — approved 10/01/2025	Review date approaching — needs annual review
Energy Management Plan (12.1)	Yes — with targets and action plan	Update with actual progress achieved
Consumption Records (12.2)	Yes — solar & grid monthly data	Need daily records, not just monthly totals
Deviation Analysis (12.2)	Partial — remarks on solar data	Need formal RCA/CAPA format
Training Records (12.3)	Yes — 14 employees trained 08/03/2025	This workshop adds another training record
Energy Audit (12.4)	Yes — checklist completed 16/04/2025	Need periodic audits (schedule quarterly)
RCA/CAPA (12.5)	Process exists but not formalized for energy	Need 2-3 documented energy CAPAs

Action Items After This Training

#	Action	Who	When
1	Start daily meter reading recording	Designated operator	Immediately
2	Calculate SEC (kWh/part) for baseline	Production + Admin	Within 1 week
3	Set quarterly energy targets (Q1 2026)	Management + Energy team	Within 1 week
4	Complete 2-3 formal RCA/CAPAs from historical deviations	Energy team + Akshaya Createch	Within 2 weeks
5	Schedule next internal energy audit	Internal auditor (Nadhiya?)	Within 1 month
6	Review and update Energy Policy (annual review due)	Sandeep G. Parvatikar	Within 1 month
7	Evaluate digital energy management system	Management + Akshaya Createch	Within 2 weeks

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Thank You

Unnathi CNC Technologies — On the path to ZED Excellence

Faculty: Lakshminarasimhan K & Aravind **Organization:** Akshaya Createch

For questions and follow-up: Contact Akshaya Createch